Electromyographic Analysis of Row Exercises

How does exercise classification impact muscle activation induced by the row movement?

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Word Count: 4996

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Introduction

Once used exclusively by elite athletes and weight lifting enthusiasts, resistance training has substantially grown in popularity over the past 3 decades, with over 162 million private gym members worldwide. (Addolorato et al., 2020; McMaster et al., 2009). This growth of the fitness industry has facilitated an amplified common interest in optimizing outcomes of resistance training, namely improvements in long-term strength and hypertrophy (LTSH). As a result, the effectiveness of commonly performed resistance exercises has been questioned (Addolorato et al., 2020). Exercise effectiveness is defined as the efficacy of producing improvements in long-term strength and hypertrophy (LTSH) when implemented in a long-term resistance training program. Exercise effectiveness can be quantified by measuring the total number of muscle motor units recruited during exercise performance, commonly known as muscle activity (Maden-Wilkinson et al., 2020). While intensity is the single most important factor in muscle activity levels, the biomechanics of an exercise also plays a significant role (Stastny et al., 2017). Biomechanics, or biomechanical principles, relate force and momentum to the human musculature, namely range of motion (ROM), time under tension (TUT), stability, and joint angle (JA) (Bennett, 2019; Martins-Costa et al., 2014; Picerno et al., 2017; Signorile et al., 2020). The impact of these biomechanical principles on exercise effectiveness has prompted the development of new exercise equipment and techniques for performing these exercises to maximize ROM, TUT, stability, and JA (Addolorato et al., 2020; Bennett, 2019).

While traditional resistance training often utilizes calisthenic exercises and free weights, modern resistance training protocols often incorporate machines and specific techniques of exercise performance designed to optimize effectiveness (Fisher et al., 2011). While machines perform the same basic movements as free weights, they often provide greater stability, ROM, and variable resistance, designed to increase exercise effectiveness. However, until the effectiveness of modern machine exercises and free weights are experimentally compared, it is unknown whether or not modern equipment is necessary to optimize improvements in LTSH in their target musculature—defined as the collection of muscle groups designed to perform a particular movement (Fisher et al., 2011; Youdas et al., 2010). Particular techniques of performing machine-based resistance exercises have also been developed to increase exercise effectiveness by further optimizing ROM, stability, JA, and movement control; these exercises can be said to be optimized for movement quality (MQ) (Bennett et al., 2019). Because free-weight variations of the row movement necessitate the high activation of stabilizing muscle groups rather than the movement's target musculature, machines are an attractive alternative that may provide greater activation of the row target musculature (Huegli, 1993).

Past literature has indirectly evaluated the effectiveness of different individual resistance exercises which perform the row movement by examining exercises that target the back. For example, Edelberg (2017) examined the muscle activity elicited by 8 resistance exercises targeting the back musculature, including 3 variations of the row, determining that no one exercise was most effective for targeting the back musculature. Past literature has also assessed the advantages provided by different techniques of performing an individual row exercise. Specifically, two variations of the seated cable row were compared—with shoulders slacked and with shoulders retracted—and no significant differences in muscle activation resulting from shoulder positioning were observed (Lehman et al., 2013). However, there is insufficient research covering muscle activation elicited by variations of the row movement. The present study aims to address this by examining the row movement in particular across a variety of exercise classifications.

The purpose of the present study was to explore the patterns between the classification of row exercise variations—namely free weight, cable and machine classifications—and levels of muscle activation within the row's corresponding target musculature. By doing so, the advantages (or lack thereof) of possessing and/or utilizing modern resistance training equipment for the row movement will be made clear. The population of apparently healthy young adult males with a minimum of 1 year resistance training experience were chosen as the population for the present study in order to minimize inconsistent and unreliable results due to injury, pre-existing health conditions, lack of experience, and gender (Stasny et al., 2017; Maden-Wilkinson et al., 2020; Mehls et al., 2022). Due to Edelburg's (2017) conclusion that the bent-over barbell row, a free-weight exercise, elicited the greatest overall muscle activation in the back musculature, which overlaps greatly with the row target musculature, it was hypothesized that the dumbbell row, a similar free weight exercise, will elicit the greatest level of muscle activation within the row target musculature.

Literature review

Electromyographic analysis of exercise classifications

Exercise classifications, defined using the biomechanical principles optimized by the exercise and equipment used (e.g free weights, conventional / isokinetic machines), can be evaluated as a whole based on how effectively they activate the target musculature (Aboodarda et al., 2016). Stabilization required by an exercise classification, for example, can have a substantial negative impact on exercise effectiveness. When considering pressing movements from a standing position, namely the standing cable press (SCP), the mechanics are significantly different from traditional supine pressing movements like the bench press (BP) (Santana et al., 2007). Consequently, high activation of stabilizing muscles sets a limit of activation of primary movers, and thus, limits strength and hypertrophy of the target musculature. This conclusion extends to pulling movements similar to the row; chin-ups were found to elicit greater biceps brachii and erector spinae activation than the lat pulldown, indicating that the lat pulldown targets the latissimus dorsi to a greater extent, but creates overall less muscle activation than the chin-up (Doma et al., 2013). Such a result is substantiated by greater necessity for stabilization in the chin-up, causing stabilizing muscle groups (e.g. rectus abdominis) outside the target musculature to contract. While the SCP and chin-up are classified differently (cable and calisthenic, respectively), and have different target musculature, their lack of stabilization decreases activation of the target musculature. Therefore, greater stabilization during performance of the row movement is likely to promote greater muscle activation and thereby superior improvements in LTSH within the row musculature.

Although it is possible to evaluate exercise classifications indirectly by identifying patterns in studies comparing individual exercises, exercise classifications as a whole can be directly evaluated and compared. For example, elastic resistance and isoinertial resistance exercises have been found to be comparable in their ability to provide muscle stimulus, and the adaptability of cable exercises is typically favored over free weights to improve movement quality (MQ) (Aboodarda et al., 2016; Bennett et al., 2019). Although the improved MQ was not found to result in greater muscular stimulus, adapted exercises produced similar strength and physical performance outcomes as traditional resistance training at a lower perceived training intensity (Bennett et al., 2019). The contrast in effectiveness between exercise classifications based on their biomechanics suggests that the row is also subject to variability in muscle activation based on the different variations of the row. The present study aims to explore this contrast in effectiveness between classifications of the row, a movement not previously studied, and establish the reasons for this contrast (Aboodarda et al., 2016; Bennett et al., 2019).

Electromyographic analysis of back exercises

While few previous studies have evaluated the effectiveness of exercise classifications for the row movement in particular, past research has evaluated row exercises indirectly by examining a range of exercises targeting the muscles of the back. For example, eight different exercises targeting the back musculature across several movements (e.g. pulldown, row, rear fly) were evaluated and compared to discover the best exercise for targeting the back musculature (Edelburg, 2017). No single exercise emerged as most effective at targeting the back musculature; rather, a combination of exercises was found to induce optimal improvements in LTSH. Further, there was high variability between the select classifications of row exercises examined (i.e. calisthenic, free weight), with the bent-over row producing the highest overall muscle activation among all exercises and the TRX row producing one of the lowest (Edelburg, 2017). Thus, when examining exercises spanning several classifications which target the back musculature, muscle activation (and thereby exercise effectiveness) heavily varies between exercises and classifications, especially when examining exercises which emulate the row movement. Additionally, the wide grip pull down, reverse grip pull down, seated row with shoulders retracted, and seated row with shoulders slack, were evaluated electromyographically (Lehman et al., 2004). While the performance techniques of these exercises differed, biceps brachii activation did not differ significantly, and retracting the scapula did not appear to have a significant effect on the level of trapezius/rhomboid activation. Back exercises of the same classification, regardless of performance technique, are therefore likely to produce comparable levels of muscle activation. Thus, conclusions made on the effectiveness of a single variation of the row is likely to hold true for all row exercises which share the same classification. In conjunction, the conclusions of variability in effectiveness between classification of row exercises and consistency within these classifications indicates that comparing a single row exercise for each classification will give accurate insight into the most effective classification as a whole for the row. The present study aims to use this reasoning to determine the most effective exercise classifications in facilitating improvements in LTSH within the row movement's target musculature, and thereby the equipment necessary to optimize improvements in LTSH.

Past research has established differences in effectiveness between row exercises and their classifications, and evaluated individual row exercises with respect to other exercises targeting similar musculature. However, the most effective exercise(s) for the row movement remains unclear.

Method

The purpose of the present study was to explore the patterns between the classification of row exercise variations and levels of muscle activation within the row's corresponding target musculature. Electromyographical analysis allows muscle activation levels to be clearly quantified, and thus compared between variations of the rowing movement spanning several exercise classifications. The specific row exercise(s) which elicits the highest overall muscle activation level for the most muscle groups can thereby be determined to be the most effective between all examined classifications. Hence, the advantages (or lack thereof) of possessing and/or utilizing modern resistance training equipment for the row movement will be made clear.

Surface Electromyography

To compare the relative effectiveness of various row exercises, it is necessary to quantify the efficacy of an exercise in producing improvements in long-term strength and hypertrophy (LTSH). This quantification can be achieved by the means of surface electromyography (EMG), measuring muscle activity in units of volts, to achieve a quantity proportional to the level of muscle activation. A positive, linear relationship has been established between muscle activity levels and improvements in both strength and muscle hypertrophy (Maden-Wilkinson et al., 2020). Further, surface electromyography (EMG) has been shown to be the most accurate tool for the measurement of muscle activity (Hill et al., 2019). Thus, surface EMG was the most accurate tool available to evaluate the effectiveness of each exercise in producing improvements in long-term strength and hypertrophy. The peak amplitude of each electrical impulse (in Volts) represents the relative number of motor units recruited in the muscle group, functioning as a measurement of total muscle activation for each muscle group (Knutson et al., 1994). Patterns can thereby be established between the level of muscle activation during performance of a particular exercise and the exercise classification(s) the given exercise belongs to. Both surface EMG equipment and resistance exercise equipment were made available by the nutrition and kinesiology department of a local university.

Population

The participant base included 18 year old high school students involved in non-professional weightlifting activities. Potential participants were identified according to their status of 6 months or more of weight training experience, male sex, lack of apparent injury, and accessed via an informal personal survey at a local high school and large, commercial gyms in Chicago. This population of apparently healthy, young adult males was chosen because they are least likely to suffer from injuries and health complications which may impact exercise performance (Edelburg, 2017). The existence of injuries and/or health complications in a subject is highly likely to diminish the intensity at which exercises can be performed, thereby substantially reducing the reliability of results (Stastny et al., 2017). Further, strictly males were chosen to eliminate any confounding variables, namely sex, due to inconsistencies in EMG-measured muscle activation levels between sexes (Mehls et al., 2022). Subjects also had at least 1 year of previous resistance training experience so that they were familiar with the lifts being conducted in the current study (Edelburg, 2017; Stastny et al., 2017). Each participant was informed of the purpose, procedures, potential risks, and benefits of the study before providing written consent for voluntary participation via a consent form (Appendix A) adapted from Edelburg (2017).

Procedure

The present study examined three unilateral row exercises: Single-Arm Dumbbell Row (DBR), Single-Arm Stiff Machine Row (MR), and Single-Arm Cable Row (CR). DBR involved standing with one foot on the ground, bending over with arm fully extended, and pulling the dumbbell perpendicular to the torso towards the hip (Graham, 2001). MR involved sitting on a machine and pulling the handle towards the hip. CR involved sitting on a bench facing the cable-cross machine, pulling the handle towards the hip, and releasing it back to the starting position (Ronai, 2019).

Each subject was required to attend one familiarization session, performing each exercise under the guidance of the principal investigator in a non-measured setting prior to data collection in order to demonstrate familiarity with proper form for each exercise and thereby ensure reliability of EMG data (Edelburg, 2017). During this session each participant’s 1 repetition maximum (1RM) was established by gradually increasing the weight for each exercise in order to determine the maximum weight with which 8 reps can be performed: approximately 80% of 1RM according to the Epley formula (Epley, 1985). With a minimum of 48 hours rest following the familiarization session to maximize ATP recovery and minimize fatigue during testing, subjects then completed one testing session (Edelburg, 2017).

For the EMG measurements, four self-adhesive electrodes were placed on the skin, over muscles of the upper back, lower back, and arms. The individual muscles examined were the latissimus dorsi (LD), biceps brachii (BB), posterior deltoid (PD), and middle trapezius (MT). Electrodes were placed according to Figure A for subjects with right-handed dexterity, and placement was mirrored across the body for subjects with left-handedness (Botton et al., 2013; Edelburg, 2017; Lehman et al., 2013). Electromyographic (EMG) signals were recorded at 1200 Hz (Motion Labs, Baton Rouge, LA) using bipolar surface electrodes (Norotrode 20, Myotronics – Noromed, Inc., USA).

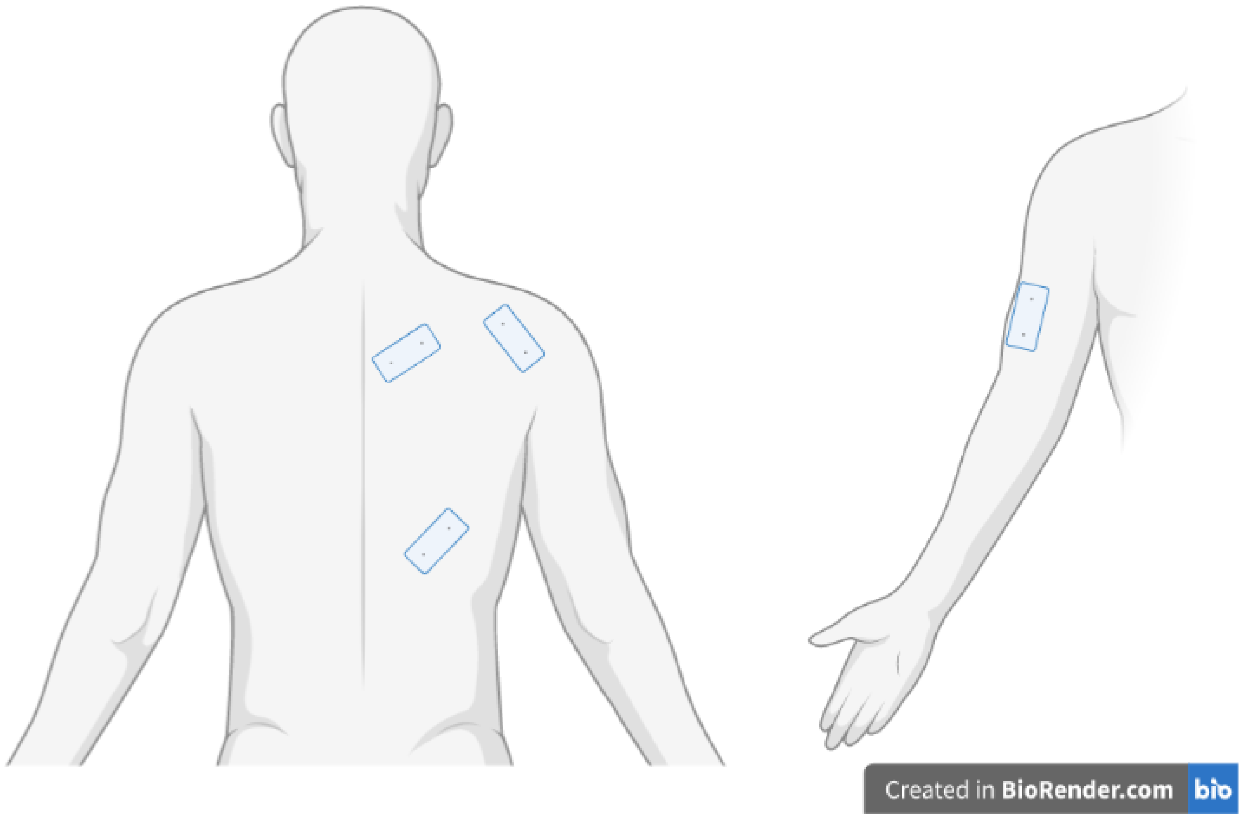

Figure A: Electrode Placement

The testing session consisted of performing these three exercises, in a randomly assigned order, using 80% of their 1RM. 80% of each participant's 1RM for each exercise was used due to optimal reliability in EMG data when an exercise is performed at highest possible intensity, for approximately 8 repetitions, and as close to, without exceeding 80% of a participant's 1RM (Picerno, 2017; Stastny et al., 2017; Youdas et al., 2010). In between each exercise set, the subject rested for a minimum of 3 minutes to ensure ATP recovery and reliability of a set's mean rep kinetic data over data from the best performed rep (Picerno, 2017). This process repeated until all three exercises for each participant were examined.

Normalization, Signal Processing, and Statistical Analysis

To effectively compare the effectiveness of exercises across subjects, it is necessary to normalize EMG amplitudes according to each subject's capabilities to account for strength and neurological differences between subjects (Sousa & Tavares, 2012). According to the normalization matrix contracted by the CEDE project, the method of dynamic MVC normalization was chosen; in this case, normalization to peak voltage of a single exercise condition (Besomi et al., 2020); Yang & Winter, 1984). The single exercise that was chosen for the EMG signals to be normalized to was the dumbbell row (DBR) due to its high level of repeatability and widely standardized nature (Stone, 2000). This method of normalization involves recording raw EMG data (in Volts) for each muscle group during the DBR, and normalizing all EMG signals for each muscle group to that muscle groups global maximum voltage during the DBR (Yang & Winter, 1984). The resulting EMG data will be expressed as a percentage of the peak voltage recorded during the dumbell row for that muscle group (%PDBR). The peak normalized EMG value for each of the 8 repetitions is thereupon identified and averaged across all 8 repetitions to obtain an average peak EMG value for each muscle group and each subject. This method was determined to be the most feasible and highly effective for comparing activation in the same muscle group across several exercises (Besomi et al., 2020; Sousa & Tavares, 2012).

To evaluate and compare the effectiveness of each row exercise, an EMG signal processing and statistical analysis framework to average normalized muscle activity levels across subjects and compare exercises was necessary (Hermens et al., 2000). EMG signals were: i) band-passed filtered (20-400Hz) to remove non-muscle frequencies, ii) high-pass filtered (50Hz) to remove motion artifact, iii) notch-filtered (59.5-60.5Hz) to remove ambient power line noise

(Seyedali et al., 2012; Huang & Ferris, 2012) iv) full wave rectified, v) low-pass filtered (10 Hz Butterworth) to obtain a linear envelope of each EMG signal, and vi) amplitude normalized to the peak of each participant's DBR condition (Yang & Winter, 1984).

Normalized EMG activity between exercises was compared using a paired sample t-test for means. This individual comparison between exercises was repeated 3 times for each muscle group and for the averaged EMG across all muscle groups. Alpha was set at 0.05 to achieve statistical significance (Edelburg, 2017). Signal processing and statistical analysis software was made available by the Kinesiology department of a local university.

Results

Demographics and Anthropometric Measurements

Six physically active, young adult male subjects with no conflicting injuries or health complications participated in the present study. Demographics and anthropometric measurements of the subjects are presented in Table 1 and 2, respectively. All subjects who participated in the present study were 18 years of age and male, with a minimum of 1 year experience resistance training (Table 1). On average, subjects were 176.50 cm tall and weighed 80.78 kg (Table 2).

Table 1*: Participant Demographics (n=6)*

Characteristic		n
Gender		
	Male	6
	Female	0
Age		
	18	6
Experience level		
	1-2 years	1
	2-3 years	2
	3-4 years	2

>4 years	1

Table 2*: Subject Anthropometric Measurements (n=6)*

Characteristic	Mean	SD	Range
Height (cm)	176.50	6.16	166-185
Weight (kg)	80.78	8.13	69.6-88.3

Electromyographical Analysis

The electromyography (EMG) data collected for this study was analyzed to determine the level of muscle activation produced by three different exercises: dumbbell rows (DBR), cable rows (CR), and machine rows (MR). The purpose of this study was to compare the muscle activation patterns between the exercises and determine which exercise is most effective for overall muscle activation.The results of the present study regarding the normalized muscle activation (%PDBR) for each of the four muscle groups during each exercise are presented in Table 3.

When compared with the CR and DBR, it was found that the MR elicited the most muscle activation on average. This can chiefly be attributed to the finding that, when muscle activation (in %PDBR) was averaged across all examined muscle groups, the MR elicited significantly more muscle activation than the DBR with a difference in mean muscle activation of 14.03% and a P value less than 0.05 (Table 3). When examining individual muscle groups, significantly higher overall activation during the MR can primarily be attributed to the BB, where the MR achieved a mean muscle activation of 96.56%, compared to 75.49% and 65.40% during the DBR and CR, respectively. Although MR activation was not significantly higher than the DBR in the BB, it was significantly higher than the CR at a P value less than 0.01. Furthermore, the MR also was found to elicit significantly higher muscle activation than the

DBR in the PD muscle group at a mean difference of 10.66% and a P-value less than 0.05. All results finding a positive mean difference between the MR and both alternative row variations were met with at least one statistically significant finding, and all statistically significant findings pertained to higher activation during performance of the MR when compared to either the DBR or the CR.

Although all statistically significant results pertain to the MR eliciting higher muscle activation than either the MR or the CR, descriptive statistics for the LD muscle group are worth noting. Here, the mean activation for the CR was 125.01%, compared to 105.65% and 78.32% for the MR and DBR, respectively (Table 3). A large standard deviation during CR performance (86.27%) prevents statistically significant results for the CR and MR, but the mean difference between the CR and DBR, 46.69%, was the highest between variations for any examined muscle group. This caused the CR to elicit the second highest muscle activation, on average, between all row variations, at 85.26% PDBR (Table 3).

The least notable differences were observed for the MT muscle group. Here, the DBR elicited the highest muscle activation at 72.02% PDBR, but obtained only a 2.94% and 3.37% mean difference between itself and the MR and CR, respectively (Table 3). No significant differences were found for the MT. This caused the DBR to elicit the least muscle activation, on average, of all examined row variations, at 74.55% PDBR (Table 3).

Interestingly, the variability in muscle activation between the three exercises was most significant in the BB and LD, while relatively similar between exercises for the MT and PD (Table 3). As previously mentioned, muscle activation was particularly variable during CR performance, as exemplified by the standard deviation of 86.27% in muscle activation between subjects for the LD during CR performance. Additionally, the DBR elicited the least variable muscle activation between subjects across all muscle groups, while the CR activation was the most variable. Overall, all statistically significant results indicated greater activation during the performance of the MR when compared to the DBR or CR.

Table 3: *Average peak EMG (%PDBR) across all muscle groups during the 3 exercises*

Exercise	DBR		MR		CR	
EMG Statistic	Mean	SD	Mean	SD	Mean	SD
BB	75.49%	16.44%	†96.56%	36.10%	65.40%	23.82%
LD	78.32%	13.11%	105.65%	58.57%	125.01%	86.27%
MT	72.02%	20.32%	69.08%	24.98%	68.65%	14.77%
PD	72.36%	17.44%	*83.02%	17.43%	81.99%	20.79%
Overall	74.55%	4.21%	*88.58%	18.21%	85.26%	25.34%

*Significantly greater activation than DBR ($P < 0.05$)
†Significantly greater activation than CR ($P < 0.01$)

Discussion

The results of this study reject the primary hypothesis that free weight row exercise variations elicit the highest level of muscle activation within the row target musculature; instead, results support the alternative hypothesis that conventional machine row variations elicit the most. The present study was the first in its field to focus on the row movement; in particular, how exercise classification (free-weight, conventional machine, cable) impacts muscle activation in the movement's target musculature. The findings of present demonstrate that exercise classification has a significant impact on muscle activation patterns and overall levels in the row target musculature. Specifically, i) machine classifications tend to elicit significantly more overall activation than free weight classifications, ii) machine classifications tend to elicit significantly more activation in the PD than free weight classifications, and iii) machine classifications tend to elicit significantly more BB activation than cable variations.

Most notably, when normalized muscle activity levels were averaged across all muscle groups, the MR was shown to exhibit significantly more overall muscle activation than the DBR, but not significantly more than the CR. When comparing individual muscle groups, the MR was shown to elicit significantly higher PD activation than the DBR variation, and significantly higher BB activation than the CR variation. These results primarily suggest the superiority of the MR over the DBR in terms of eliciting the greatest amount of muscle activity and therefore also in terms of maximizing LTSH. This conclusion can be extrapolated to suggest the superiority of conventional machine exercises over free weight exercises in terms of maximizing LTSH within the row target musculature, specifically for unilateral movements. This finding opposes previous studies concluding the general superiority of free weight exercises over conventional machine and cable exercise variations for maximized outcomes in long-term strength and hypertrophy (LTSH) (Edelburg, 2017; Stone et al., 2000). However, this finding supports studies advocating stabilized movements over unstable movements for maximizing muscle activation within an exercise's target musculature, given that the MR provides significantly more stability than the DBR (Santana et al., 2007).

In general, previous studies have often concluded that free weight exercises, performed correctly, are superior to conventional machine and cable exercises in terms of the level of muscle activation and therefore outcomes in long-term strength and hypertrophy (LTSH) (Santana et al., 2007; Schwanbeck et al., 2009; Stone, 2000). This has been found to be especially true for compound movements, movements which require the activation/utilization of several major muscle groups, such as the squat (Schwanbeck et al., 2009) and the bench press (Santana et al., 2012). As a result, it is unexpected that a similar compound movement with a different target musculature, the row, elicits the most activation when performed using a conventional machine variation. A possible explanation for this finding is the lack of stabilization muscles within the row target musculature (e.g. rectus abdominis), particularly for the four muscle groups examined in this study (BB, LD, MT, PD). As a result, exercises which require stabilization (DBR) from muscle groups outside this examined musculature, e.g. rectus abdominis, may limit activation of the target musculature, especially considering the unilateral nature of the exercises performed in this study. Exercise variations which provide greater stability (MR) would therefore stimulate greater activation and improved LTSH in the target musculature.

Moreover, in studies examining a particular target musculature (in this case the back musculature) rather than a particular movement, the bent over barbell row, a free-weight exercise was determined to be the most effective in activating the back musculature (Edelburg, 2017). A possible explanation for the differing results in this study, which examined the row movement rather than the back musculature, is the inclusion of BB in the EMG analysis (a muscle group separate from the back musculature). The most statistically significant ($P < 0.01$) result of the present study indicates that the MR elicited significantly higher biceps brachii (BB) muscle activation than the CR; this finding suggests the superiority of the MR over the CR when attempting to optimize LTSH within the biceps musculature. When examining overall muscle activation as a result of the row movement, the BB acts as a primary mover; although the MR was not shown to elicit significantly higher BB activation than the DBR in particular, BB activation levels observed to be generally greater during MR performance in the present study likely account for the significantly higher overall activation levels in a conventional machine row variation when compared to a free weight variation.

Limitations

The most apparent limitation of this study is the small sample size. Although the time intensive one-on-one procedure familiarization with participants, relatively air-tight data collection, and data normalization ensure the reliability of results and extrapolation to the larger population of all amateur weightlifters, variability in muscle activation (most apparently in the LD) between subjects is still observed. Moreover, the maximum load capabilities of the available exercise equipment exist as the secondary limitation of the present study. Because of the high neuromuscular capabilities of the more experienced subjects (>3 years resistance training experience), a small number were able to perform 8 repetitions of the CR at the maximum load allowed by the available cable-cross machine. Although each subject's true 8RM load value did not exceed the maximum weight by more than 10%, it still affects the primary determinant of LTSH, intensity, and can likely account for the high variability in LD activation for the CR between participants. However, this limitation does not impact the primary finding of this study, that the MR is superior to the DBR in terms of eliciting more overall muscle activation and maximizing LTSH across the row target musculature as a whole. Because of the indication that the CR may be much more effective for LD activation by descriptive statistics, further research is needed to determine the effectiveness of the CR in eliciting LD activation in comparison to free weight exercises such as the DBR.

Conclusion

In summary, the present study demonstrates that modern strength training equipment, specifically conventional machines, offers a significant advantage over free weights in optimizing row training protocols for LTSH, particularly in unilateral exercises. Additionally, the study suggests that cable row exercises may yield better LTSH outcomes compared to free-weight exercises; however, further research is required to confirm this.

Although the methodology, data collection, and analysis employed in this study were rigorous and well-executed, it is crucial to acknowledge that the results may not be entirely generalizable. The general population might not accurately represent every athlete, as resistance training objectives can vary significantly among individuals. Factors such as genetics, diet, long-term program consistency, and the unique skeletal geometry and musculature of the athlete greatly influence LTSH outcomes. Nevertheless, the conclusions derived from electromyographical data provide valuable insights into the intrinsic advantages of different resistance exercise equipment concerning human neuromuscular physiology and biomechanical principles. This information is essential because it can inform the development of more effective training programs that can benefit athletes and fitness enthusiasts seeking to maximize their LTSH.

Given the potential advantages of cable equipment in performing unilateral row movements suggested by descriptive statistics, future research should further explore the efficacy of cable row equipment. Specifically, future studies should compare muscle activation between unilateral cable row variations and free weight variations, with a focus on the latissimus dorsi (LD) muscle group. Moreover, it is crucial for future research to consider the advanced neuromuscular capabilities of experienced subjects and utilize cable-cross equipment that allows for higher resistance levels (greater than 150 pounds of resistance per weight stack).

Despite its limitations, the current study contributes valuable new knowledge to the field of resistance training by providing compelling evidence for the benefits of using modern resistance training equipment, such as the Cybex Eagle row machine, when aiming to maximize

LTSH in the row target musculature. This information is critical for gym owners, physical therapists, coaches, trainers, and athletes, as it can guide their decisions regarding equipment purchases and training protocols to optimize LTSH in rowing-based activities, such as crew or kayaking. Furthermore, these findings on cable row exercises could potentially shape the development of more effective training programs in the future. Ultimately, this study emphasizes the importance of incorporating modern resistance training equipment into training protocols to maximize LTSH for athletes and fitness enthusiasts alike.

Appendix A: Consent Form

Informed Consent

ELECTROMYOGRAPHIC ANALYSIS OF ROW VARIATIONS

I, _____________________________, volunteer to participate in a research study conducted at X.

- ***Purpose and Procedure***
    - The purpose of this study is to compare muscle activity (as measured by surface EMG) during different row muscle exercises in men
    - My participation in this study will involve two sessions, each lasting approximately 1 hour each
    - The exercises to be tested will be the bent-over barbell row, dumbbell row, fixed machine row, and single-arm cable iliac row
    - During the first session, I will perform all exercises to become accustomed to them. I will also perform all exercises to establish my maximal strength (1 RM) for each lift.
    - During the second testing session, I will perform each exercise for approximately 8 repetitions, using 80% of 1 RM
    - During the two sessions, I will have adhesive electrodes placed on my upper back, lower back, and arms in order to measure muscle activity
    - I acknowledge that I may need to remove garments of clothing on my torso in order to place electrodes properly
    - Testing will take place in the weight room located in the UIC Integrative Physiology Laboratory (IPL), at 1640 W Roosevelt, Chicago, IL, 60608
    - Research will be conducted under the supervision of Dr. Andrew Sawers, a professor in the UIC Department of Kinesiology and Nutrition.
- ***Potential Risks***
    - Muscle fatigue, muscle soreness, and “pulled” muscles are possible risk factors associated with participating in this study
    - Skin irritation from placement of the EMG electrode is possible
    - Individuals trained in CPR and Advanced Cardiac Life Support will be present for all testing sessions and the test will be terminated immediately if complications occur
    - The risk of serious or life-threatening complications, for healthy individuals, like myself, is near zero
- ***Potential Benefits***
    - I, and other athletes, may benefit by gaining knowledge about which is the most effective row exercise
    - I, and other consumers of resistance training equipment, may benefit by gaining knowledge about the necessary equipment to maximize muscle activity for the rowing movement
- ***Rights & Confidentiality***

- My participation is voluntary
- I can withdraw from the study at any time, for any reason, without penalty
- The results of this study may be published in the scientific literature or presented at professional meetings using group data only
- All information will be kept confidential through the use of number codes
- My data will not be linked with personally identifiable information

I have read the information provided on this consent form. I have been informed of the purpose of the test, the procedures, and expectations of myself as well as the testers, and of the potential risks and benefits that may be associated with volunteering for this study. I have asked any and all questions that have concerned me and received clear answers so as to fully understand all aspects of the study.

If I have any other questions that arise I may contact the principal investigator: Evan Yeager (773) 251-8141

Participant:____ Date: ____________
Investigator:____ Date: _____________

Appendix B: EMG Data Collection Sheet

Subject Information

Study Name: **Electromyographic Analysis of Row Exercises**

Researcher: **Evan Yeager**

Date: ____________

Subject Code: ____________

Age (years): ____________

Sex: Male

Subject Health Information: ____________

____

____

Subject Resistance Training Familiarity: ____________

____

____

Special Notes: ____________

____

____

Roles & Responsibilities Assignment:

1. EMG Electrode Prep and Placement:

   Name: ____________________________

Subject Anthropometric Measurements

Measurement Name	**Measurement Description**
Mass (kg)	Weight of subject. Unit conversion 2.2 lbs = 1 kg
Height (cm)	Height of subject. Unit conversion 1 in. = 2.54 cm

EMG Gains

Transmitter A			
Tethered or telemetry:			
Frequency:			
Channel	**Muscle**	**Cord**	**Gain**
1	BB		
2	LAT		
3	TRAP		
4	PDELT		
Ground #:			

EMG Gain Scale

POS	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
GAIN	**0.350K**	**2K**	**4K**	**5.7K**	**8K**	**9.5K**	**11.5K**	**13.2K**	**16.6K**	**18K**

Markers:
Trial Record ("Trial01")

Trial #	**Trial Description**	**Data Collection**			
		Time (sec)	*Reps*	*Resistance*	*Notes*

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